

Salim Terzout

226-922-9759 | sterzout@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

Western University

Honours Specialization in Computer Science, Minor in Software Engineering

London, ON

Sept. 2022 – June 2026

EXPERIENCE

Software Automation Test Developer (Co-op) | C#, Selenium, SQL, Azure DevOps

May. 2026 – Present

Carfax

London, ON

- Designed, developed, and maintained **100+** automated test scripts in **C#** and **Selenium WebDriver**, validating web applications and end-to-end data integrity for every software product released to production.
- Engineered **parallel test execution** within **Azure DevOps CI/CD** pipelines, cutting automated regression suite runtime by up to **75%** and accelerating release velocity.
- Integrated production databases into test repositories, writing **SQL** queries to validate real vehicle data and third-party **API endpoints** against backend records, and refactored test architecture into reusable common classes and **NuGet**-packaged components, ensuring **100%** data accuracy and improved suite maintainability.
- Presented test findings, defects, and root-cause analysis to the **development team and QA managers**, clearly communicating bugs and product issues to support timely resolution and release decisions.

Developer - Intern | React, Python, SQL, REST APIs

May. 2025 – Present

RBC

Toronto, ON

- Built a full-stack **React** application leveraging **APIs and SQL** to retrieve and visualize **1,000+** server and portfolio records while using **pagination** and **optimized queries** to ensure fast and responsive user interaction for technical directors.
- Developed **Python** automation scripts to validate and classify large server datasets exceeding **1,000** entries, automatically flagging valid, invalid, or incomplete records, and **converting file formats** for seamless integration and reporting, eliminating manual review bottlenecks and improving data accuracy.
- Drafted clear, actionable purpose statements and next-step objectives for VPN and server management initiatives.

Software Developer | Python, Spotify API, Machine Learning

Sept. 2023 – Present

Western Cyber Society

London, ON

- Created an application that generates a playlist of **50+** songs based on user input, ensuring smooth transitions.
- Trained a machine learning model using **Spotify API** features from **10,000+** tracks to predict optimal next-track transitions

PROJECTS

Walls | React, TypeScript, Vite, Supabase, Framer Motion, Spotify API

- Built a web-first platform using **React, TypeScript, and Framer Motion** while maintaining 100% type coverage (strict mode) across **55+** files to deliver fluid **60fps** artist visualization animations.
- Optimized API performance by **50%** by refactoring sequential data fetching into **parallelized Promise.all executions**, and architected secure **OAuth 2.0 authentication** via **PKCE with maximum-entropy code verifiers**, with real-time global user matching via **Supabase database subscriptions**.

Rocket Health Monitoring System | Python, scikit-learn, ML Evaluation Metrics

- Implemented and evaluated multiple anomaly detection models for rocket telemetry data, including **Isolation Forest, Local Outlier Factor (LOF), One-Class SVM, Z-Score, and EWMA**, benchmarking each on **precision, recall, and F1-score** for point anomalies.
- Tuned SVM and LOF hyperparameters to **meaningfully improve F1-score** over baseline models, identifying the most reliable detectors across benchmarked approaches.

Diabetes Predictor | Python, scikit-learn, XGBoost, pandas, numpy

- Applied core machine learning principles to build and evaluate predictive models for diabetes risk using a large-scale healthcare dataset of **100,000** patient records with **31 health indicators**, achieving **92%** accuracy and **0.94 AUC** through an end-to-end pipeline with preprocessing, feature engineering, and feature selection identifying predictors such as **HbA1c and glucose** levels.

ChatCentral | TypeScript, React, Node.js, Express.js, PostgreSQL, Prisma

- Managed schemas and interactions using Prisma and PostgreSQL while optimizing database queries and indexing, **reducing message retrieval latency by 30%**.
- Created a real-time chat application with a type-safe **React + TypeScript** frontend for seamless messaging and **low-latency updates**, backed by scalable **Node.js** systems (**less than 200ms average response time**).

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, HTML, CSS, C++, C#, SQL, C, Bash, Unix

Frameworks/Databases: Qt, React, EJS, JavaFX, Node.js, Express.js, PostgreSQL, MongoDB

Libraries: Pandas, Numpy, Matplotlib, Seaborn, TensorFlow, Scikit-learn, PyTorch, Flask, Selenium WebDriver

Developer Tools: Jupyter Notebook, Git, Azure DevOps, Bitbucket, Jira, Confluence, NUnit, MSTest, JUnit, VS Code, IntelliJ, Eclipse

APIs/Other: REST APIs, Spotify API, Supabase, Prisma, XGBoost, OAuth 2.0